Chosen dataset: https://archive.ics.uci.edu/ml/datasets/Cervical+Cancer+Behavior+Risk

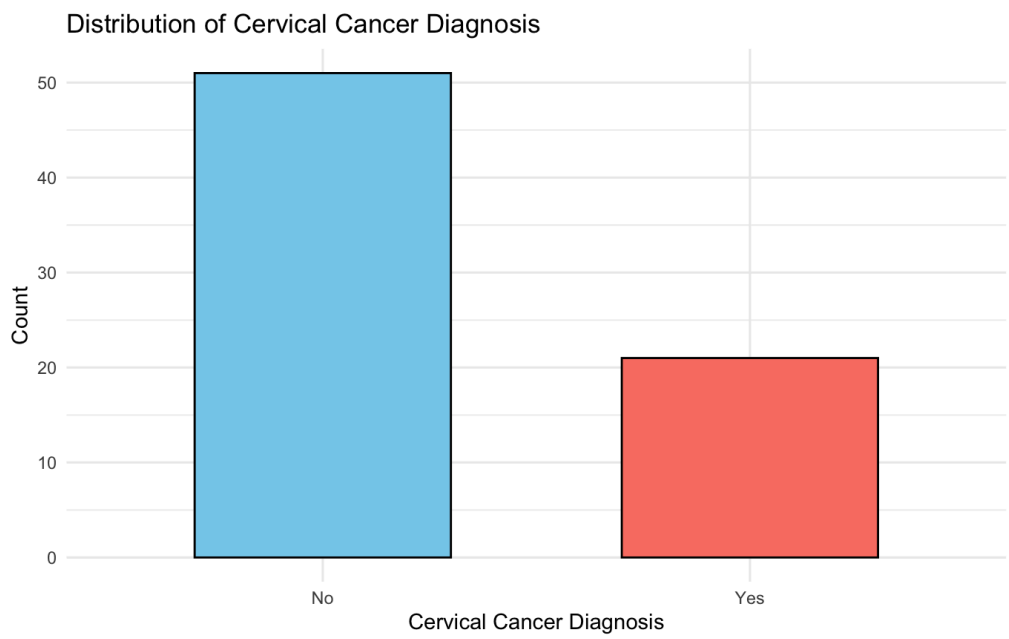
Part 1: Exploratory Data Analysis

Data Overview:

The Sobar-72 dataset has 72 patient records and 19 features with no missing data. The data points are scores from a behavioral health survey. Depending on the specific question, these scores generally range from 1 to 10 or 1 to 15. They are used to measure a patient's mindset, daily habits, and feelings of control. The 20th column is the actual diagnosis, which is strictly recorded as a 0 (no cancer) or a 1 (has cancer).

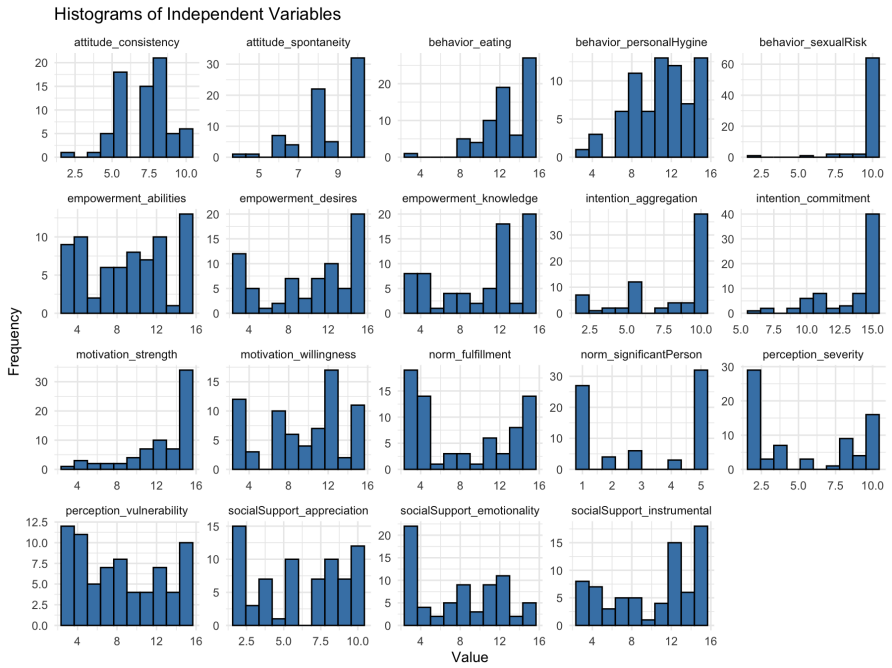
Target Variable:

The distribution bar plot below shows the classes are uneven. 51 people do not have cervical cancer, and 21 people do. Because of this imbalance, we must use metrics like Precision, Recall, and F1-Score instead of basic accuracy.

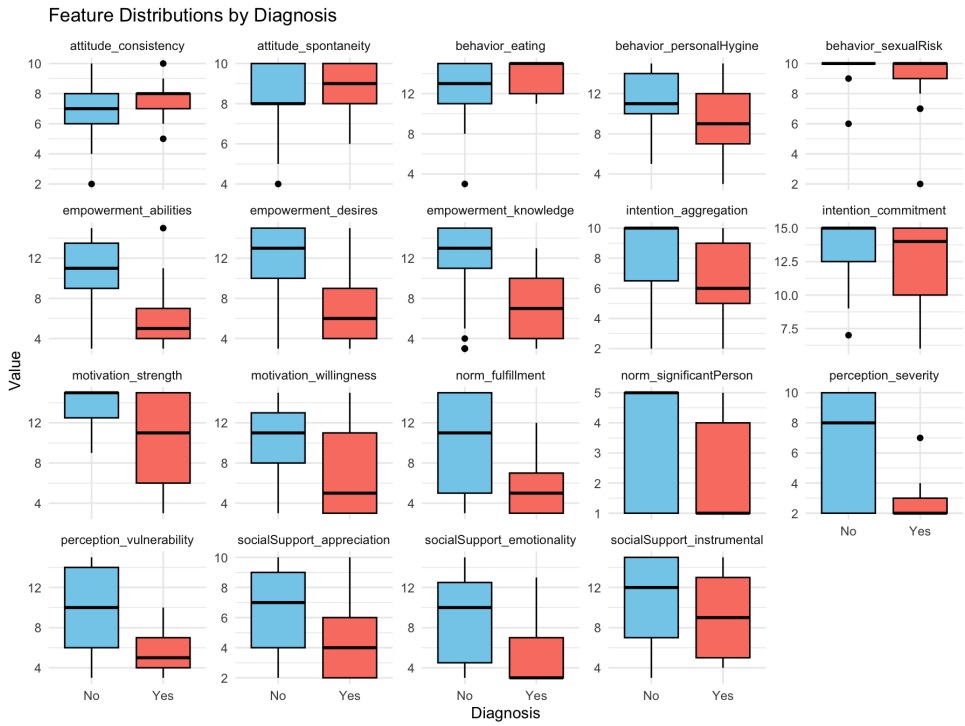


Feature Distributions and Outliers:

The histograms of independent features below show that most features are not normally distributed. They are heavily skewed. For example, behavior_sexualRisk (top right plot) is almost entirely clustered at the value of 10, while perception_severity (line 3, column 4) has a heavy leftward skew.

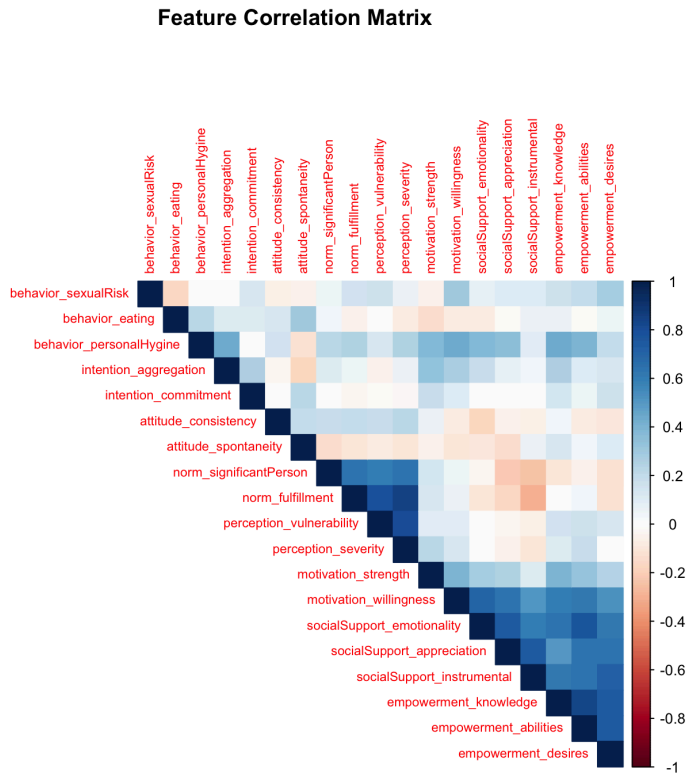


The boxplots below comparing feature distributions by diagnosis reveal even more distinction between the two classes. For example, patients with cancer exhibit lower median scores across the 3 empowerment features (empowerment_abilities, empowerment_desires, empowerment_knowledge). Despite some statistical outliers (indicated by black dots outside the whiskers), these were kept as they represent valid behavioral survey answers and because we don't have that much data to begin with.



Multicollinearity:

From the feature correlation matrix, we can see several strong positive relationships such as the empowerment features being highly correlated with one another, as are the social support features. This multicollinearity justifies the need for dimension reduction or tree-based modeling.



Part 2: Model Development, Validation, and Optimization

4 models were developed: 3 for classification and 1 for clustering. The data was split into a 75% training set and a 25% testing set. Model performance was evaluated using Precision, Recall, and F1-Score.

Model 1: Logistic Regression (Baseline Classification)

A Generalized Linear Model was fitted using all 18 features. On the test set, its precision was high at 1.0 but its recall was very poor at 0.50. This means that the model failed to predict half of the true cancer patients. This poor result is due to the small sample size and high multicollinearity.

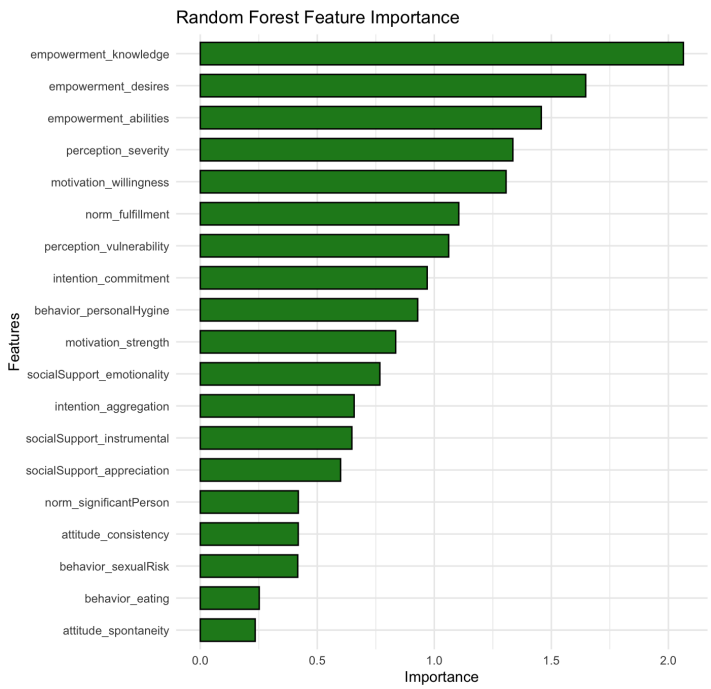
Class	Precision	Recall	F1_Score
No	0.6153846	1.0	0.7619048
Yes	1.0000000	0.5	0.6666667

Model 2: Random Forest (Classification)

To better handle the non-linear distributions and highly correlated features, I used a Random Forest, which performed better than the baseline by improving recall to 0.60 while maintaining a precision of 1.0.

Class	Precision	Recall	F1_Score
No	0.6666667	1.0	0.80
Yes	1.0000000	0.6	0.75

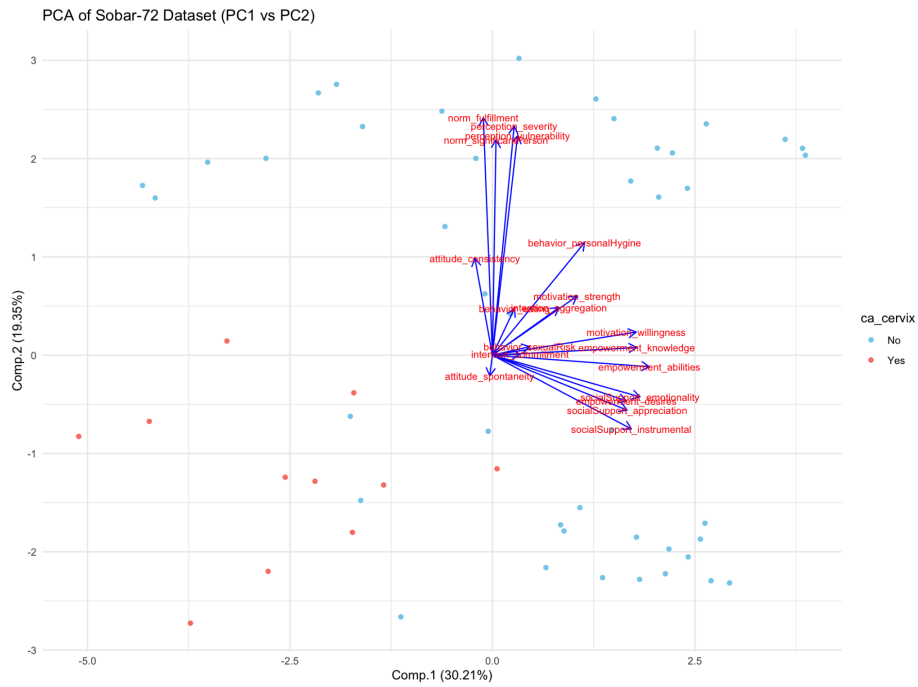
The feature importance plot also shows that the empowerment features are the primary driver of the model's decisions.



Model 3: Support Vector Machine with PCA (Optimization)

To address the multicollinearity observed in the EDA, Principal Component Analysis (PCA) was used for dimension reduction.

We visualized the projection of the data onto the first two components, with PC1 explaining 30.21% of the variance, and PC2 explaining 19.35%. From the plot, the loading vectors confirm our correlation matrix: the empowerment, social support, and motivation features all pull heavily in the same direction along the positive x-axis.

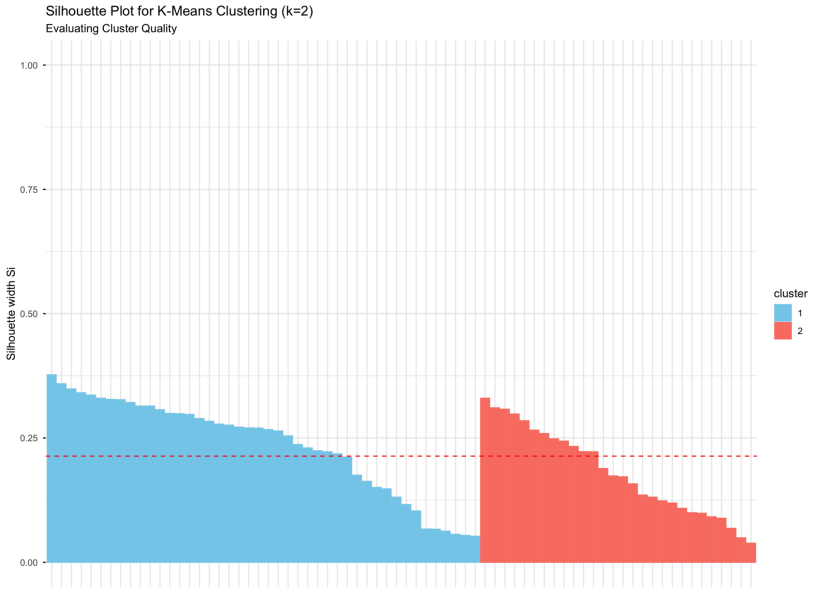


An SVM with radial kernel was then trained on the top 5 principal components, and it was the best performing model with a recall of 0.8 and a precision of 1.0 (f1-score is 0.89) on the disease positive class which is the minority class. The majority class achieved similarly high scores. We can conclude that removing overlapping features allowed the SVM to draw a much more accurate decision boundary.

Class	Precision	Recall	F1_Score
No	0.8	1.0	0.8888889
Yes	1.0	0.8	0.8888889

Model 4: K-Means Clustering (Unsupervised)

A K-Means clustering was applied to the scaled features, with k = 2 to mirror our 2 diagnostic classes. The Silhouette plot below reveals that the average silhouette width is represented by the red dashed line at 0.20, which means the clustering structure is weak. The survey data does not naturally split into two groups.



Part 3: Conclusions & Decisions

A patient's level of empowerment is a strong predictor of their cervical cancer risk. From a technical side, we proved that combining PCA with an SVM is the best way to handle small datasets with highly correlated features.

However, this model can’t be used to diagnose patients in a clinic, since the sample size is still too small (72) and the K-Means clustering proved that behavioral survey answers overlap too much to safely confirm a physical disease. Instead, since the model showed that a low level of empowerment leads to higher cancer risk, this can be used as evidence to fund wellness programs.